# ANNEXURES

## ANNEXURE 1

## PLAGIARISM REPORT

**Submission date:** 14-Apr-2026 06:39PM (UTC+0530)
**Submission ID:** 2931701218
**File name:** Project_Report_Mid_body_Final_sem_6.pdf (3.87M)
**Word count:** 15163
**Character count:** 81296

Mid

ORIGINALITY REPORT

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|---|---|---|---|
| SIMILARITY INDEX | INTERNET SOURCES | PUBLICATIONS | STUDENT PAPERS |

PRIMARY SOURCES

| | | |
|---|---|---|
| 1 | Neha J. Rao, H. Girish, M.C. Gowrishankar, Shiva Kumar, Nitesh Kumar. "A two-stage deep learning framework for kidney stone detection and clinical severity grading in CT imaging", Informatics in Medicine Unlocked, 2025<br>Publication | <1% |
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# ANNEXURE 2

# CONFERENCE PAPER SUBMISSION DETAILS

**ICAMC- 2026 Acceptance Notification for Paper ID-224**

1 message

**Microsoft CMT** <noreply@msr-cmt.org> Mon, 30 Mar 2026 at 3:09 pm
To: PushpaRaj Buvignesh <buvigneshroyal2004@gmail.com>

Dear PushpaRaj Buvignesh

Paper ID: 224
Title: Automated Parking System with License Plate Recognition and Slot Allocation

We are glad to inform you that your manuscript has been accepted for the presentation in conference ICAMC 2026 and for publication in proceeding/journal after revision.

Also, we would like to inform you that your manuscript has been reviewed and the comments from reviewers are at the bottom of this e-mail. Please incorporate the reviewers stated concerns and update the revised paper (in Camera ready paper, both doc and pdf file)

You are requested to send the following documents in attachment at ehsan.asgar@hmritm.ac.in in a Zipped file (Paper ID Number as a file name).

1. Camera Ready Paper after incorporating the reviewer comments (pdf and doc, both files). File Name: Paper ID Number_camera.doc and Paper ID Number_camera.pdf
2. Copyright Form (Paper ID Number_CTP.pdf).
   (Download from https://docs.google.com/document/d/1wxuXXoF96fJT_nVmchjqRXzyZmUeBJJp/edit)
3. Receipt of Registration Fee paid. File Name: (Paper ID Number _Fee receipt.pdf).
4. Response to Reviewers (word file). File Name: Paper ID Number _response.doc

Please follow the given template in the following link for Camera Ready Paper:
https://icamc2026.vercel.app/guidelines

Please note that the registration fee is only acceptable through the payment gateway i.e.
https://eps.eshiksa.net/DirectFeesv3/HMR_ICAMC

Conference Mode- Online/Offline (Hybrid)

The Early bird registration and to submit the required documents along with the registration fee is 5th April 2026.

Please free to contact for any query.

Looking forward for your kind response and co-operation.

Early Bird Registration Fee Details

- Students/ Research Scholar- INR 2500
- Students/ Research Scholar (for two papers)- INR 3750
- Faculty/Academician : INR 3000
- Industry Participants : INR 3500
- Attendee- INR 1000
- Foreign Participants: USD 100

Important Notes:

- Registration fees does not include accommodation and transportation expenses.
- Minimum one registration is mandatory for a paper to be the part of proceedings.

Comment to Author

Reviewer 1
1. Writing and Proofreading: The manuscript requires a comprehensive language edit to fix frequent typos (e.g., "loud-based", "automate rations") and smooth out clunky sentence structures.
2. Highlight Technical Novelty: Clearly define what makes this pipeline superior to existing integrated parking systems. If the novelty is the edge-deployment efficiency, provide explicit latency, memory, or processing metrics.
3. System Constraints: Expand the discussion on edge cases, such as how the EasyOCR module handles severely damaged plates or how the system performs under heavily degraded edge-network conditions.

Reviewer 2
Check conference template for detailed guidelines and formatting.

Best regards,

Organizing Team: ICAMC-2026

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Hamidpur, New Delhi-110036, India
Conference Email: icamc26@hmritm.ac.in
Conference Wbsite: https://icamc2026.vercel.app/
Mobile: (+91) 9999629550 (Dr. Md Ehsan Asgar, Convener)

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| Address | Chennai |
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# ANNEXURE 3
# SDG MAPPING

The project “automated parking system with number plate detection and slot allocation” is mapped to the following united nations sustainable development foals (SDGs):

SDG 9: Industry, Innovation, and Infrastructure Traditional parking and facility management rely heavily on manual ticketing, security guards, or basic proximity sensors, which are prone to bottlenecks and human error. This project modernizes physical infrastructure by introducing an innovative, automated Artificial Intelligence pipeline. By integrating state-of-the-art computer vision models—including YOLOv8 for localization, EasyOCR for text extraction, and ResNet18 for visual feature Re-Identification—the system upgrades standard IP cameras into intelligent edge devices. This demonstrates how technological innovation can be applied to build resilient, highly efficient, and scalable smart-city infrastructure.

SDG 11: Sustainable Cities and Communities Urban congestion at entry and exit chokepoints is a major contributor to localized air pollution. When vehicles are forced to stop and idle at manual parking gates, they release unnecessary exhaust emissions. By automating the authorization process through a zero-latency AI system, this project facilitates seamless vehicle flow. Drastically reducing wait times directly minimizes vehicle idling, thereby lowering the carbon footprint and localized emissions at the facility. Consequently, the project promotes cleaner air and safer, more sustainable urban transit environments within smart cities.

**Table: SDG Mapping summary**

| SDG No. | SDG Theme | Mapping (1/2/3) | Justification |
|---|---|---|---|
| 9 | Industry, Innovation, and Infrastructure | 3 | The project directly modernizes physical parking infrastructure by replacing manual ticketing with an innovative, automated Artificial Intelligence pipeline, making facility management highly efficient. |
| 11 | Sustainable Cities and Communities | 3 | By completely automating gate operations using OCR and Re-ID, the system drastically reduces vehicle wait times. This directly minimizes localized vehicle idling, lowering exhaust emissions at urban chokepoints. |

## ANNEXURE 4

## PO PSO MAPPING

Project Alignment with B.E. CSE (AI & ML) Curriculum The curriculum for the Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence & Machine Learning) is specifically designed to bridge the gap between theoretical mathematics and deployable, real-world AI applications. The "Intelligent Video Surveillance Parking System" serves as a direct, practical culmination of these educational goals.

Rather than relying on basic software development, this project requires the engineering of a highly advanced computer vision pipeline. By utilizing YOLOv8 for spatial localization, EasyOCR for text extraction, and ResNet18 for mathematical Re-Identification (Re-ID), the system solves a complex smart-city infrastructure problem. The following table illustrates exactly how the technical achievements and engineering methodologies of this project map directly to the department's mandated Program Specific Objectives (PSOs).

**Table: Program Specific Objectives (PSO) Mapping Summary**

| PSO | Mapping (1/2/3) | Justification |
|---|---|---|
| PSO 1: Apply the fundamental knowledge of mathematics, science, engineering to design and develop AI and ML solutions to solve real world problems. | 3 | The project applies mathematical concepts (Cosine Similarity vector math) and deep learning engineering to solve the real-world problem of manual |

| | | |
|---|---|---|
| | | parking gate bottlenecks and security tracking. |
| PSO 2: Build deep learning models, natural language processing and computer vision applications for healthcare, agriculture, smart city and other domains. | 3 | This project is a direct implementation of a computer vision application for a smart city domain. It successfully builds a pipeline utilizing deep learning models (YOLOv8, ResNet18) and OCR to automate facility management. |
| PSO 3: Exhibit ethical responsibility, leadership qualities, communication skills and work as a team member/leader in multidisciplinary projects. | 2 | Developing and documenting this advanced dual-verification system required strong communication of technical concepts and ethical responsibility regarding video surveillance data management and database security. |

# ANNEXURE 5
# ADDITIONAL SCREENSHOTS

The following screenshots were captured from the running Automated parking system demonstrating various features of the application.

### SCREENSHOT 1: VS VS CODE DEVELOPMENT ENVIRONMENT

The screenshot above illustrates the complete modular file architecture of the Intelligent Parking System within the Visual Studio Code IDE. To ensure maintainability and zero-latency execution, the system is divided into specialized Python modules rather than a single monolithic script:

`app.py`: The primary Flask backend server. It handles all HTTP/API requests, routes data to the web dashboard, and initializes system prerequisites upon startup.

`database.py`: The database management module. It contains the helper functions and schema initialization logic required to establish the SQLite3 connection and ensure all tables are correctly structured before the AI pipeline begins.

`final_cam_video.py`: The video ingestion module. It contains the class structures responsible for creating and managing instances of real-time RTSP video feeds from the connected IP CCTV cameras.

`pipeline_entry.py`: The master orchestration script for the entry gate. It integrates and executes all visual intelligence modules in the correct sequential order when a vehicle arrives.

`mod_vehicle_detect.py`: The initial trigger module. It actively monitors the entry/exit zones to verify the physical presence of a vehicle before activating the heavier, computationally expensive AI models.

`mod_plate_reader.py`: The Optical Character Recognition (OCR) module. Upon vehicle stabilization, this script isolates the license plate crop and extracts the human-readable alphanumeric text.

`mod_reid.py`: The deep feature extraction module. It utilizes the pre-trained ResNet18 architecture to generate the high-dimensional vector embeddings required for Cosine Similarity matching.

`mod_occupancy.py`: The spatial awareness module. It analyzes the parking bay camera feeds to identify empty slots, returning the specific zone names and coordinate data for vehicle assignment.

`mod_display.py`: The human-machine interface (HMI) logic. It generates real-time visual instructions for drivers at the gates (e.g., "Stop," "Proceed to Bay 4") to prevent miscommunication and ensure smooth facility traffic flow.

`parking_system.db`: The localized, serverless SQLite database file that securely stores all hardware configurations, parking lot spatial data, and the historical ledger of vehicle entry/exit logs.

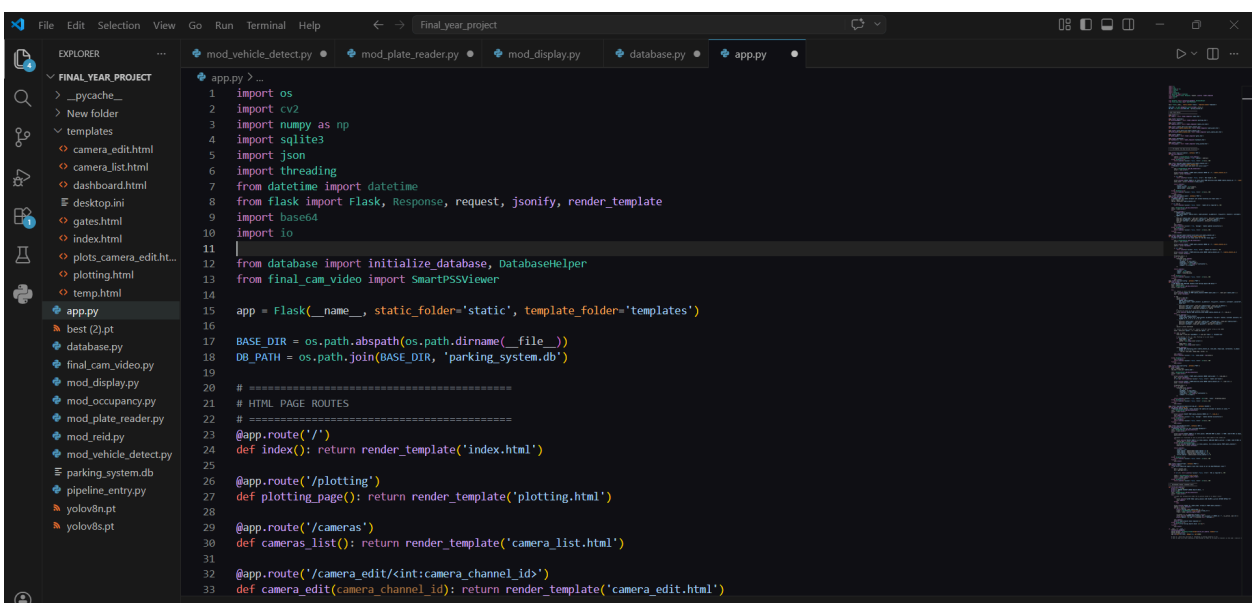


***Fig A5.1: VS Code - Project Structure***

## SCREENSHOT 2: SPATIAL CONFIGURATION AND ZONE MAPPING INTERFACE

This specialized web interface allows administrators to dynamically map physical parking bays into the system's logic without hard-coding coordinates. Key functionalities include:

Visual Digitization: Administrators draw exact mathematical polygons (such as the green 'p1' boundary) directly over the live camera feed to define parking regions.

Database Synchronization: Clicking "Save to DB" instantly translates these visual shapes into 2D spatial coordinates and logs them into the `detection_zones` SQLite table via a Flask API.

AI Integration: These saved coordinates provide the mathematical baseline for the YOLOv8 occupancy model (`mod_occupancy.py`), which calculates real-time parking availability based on boundary intersections.

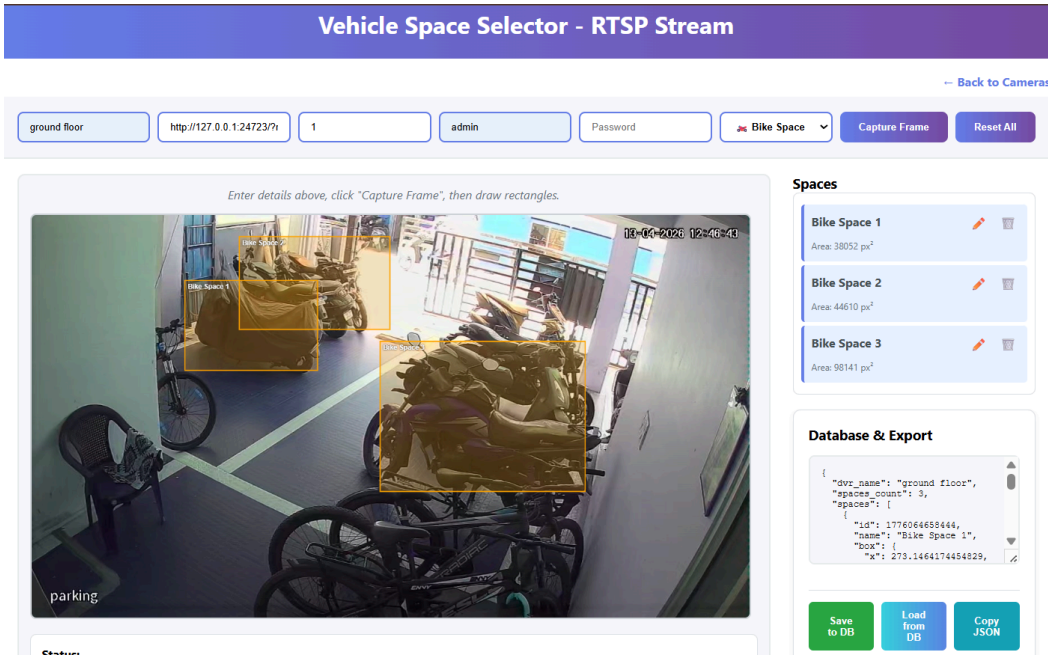


***Figure A5.2: Web-based interface for dynamic digitization and database mapping of parking space coordinates***

## SCREENSHOT 3: CAMERA & ASSET MANAGEMENT DASHBOARD

Camera Management Dashboard. This interface allows for the real-time configuration of camera stream credentials (IP/RTSP) and the management of associated parking space counts within the system database.

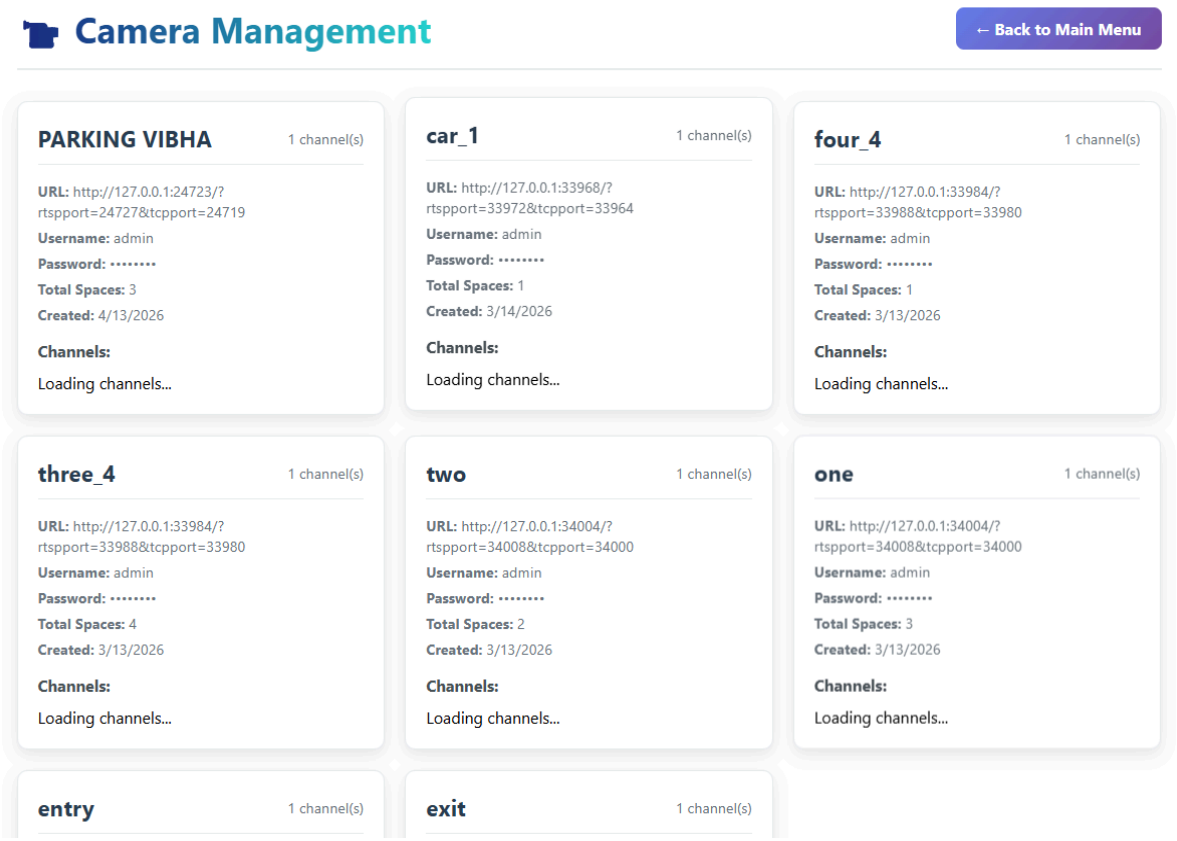


***Fig A5.3: Camera & Asset Management Dashboard***